

Electrical Properties

The Electrical properties of a material are those which determine ability of material to be suitable for a particular Electrical engineering Application. Some of the typical Electrical properties of engineering materials are listed below.

1. Resistivity
2. Conductivity
3. Temperature co-efficient of Resistance
4. Thermoelectricity

Mechanical Properties of Engineering Materials

Being an Engineer, we must have a thorough knowledge of mechanical properties of Engineering Materials, because they are of great importance in the design of tools, machines and structures. The most important and useful mechanical properties are briefly explained below.

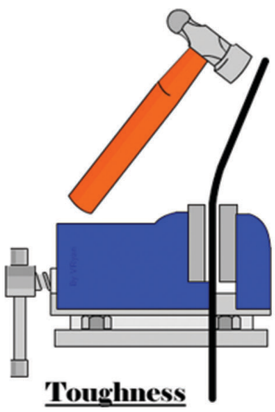
1. Toughness
2. Hardness
3. Brittleness
4. Malleability
5. Stiffness
6. Ductility
7. Elasticity

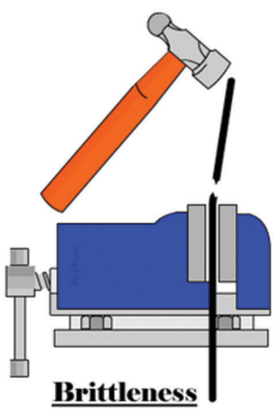
Toughness: Toughness is the ability of a material to withstand sudden external forces. It is the amount of energy absorbed by the material before it develops fracture. Wrought Iron, Mild steel are the examples for the tough materials.

Brittleness: Brittleness of a material indicates that how easily it gets fractured when it is subjected to a force or load.

Example Glass, Cast Iron.

Hardness: Hardness is a fundamental properties which is closely related to the ability of a material the resist to scratching, abrasion, and cutting. High carbon steels and high speed steels have higher hardness. Diamond is the hardest material.







GM_Chp04.indd 56 14-02-2019 11:35:58